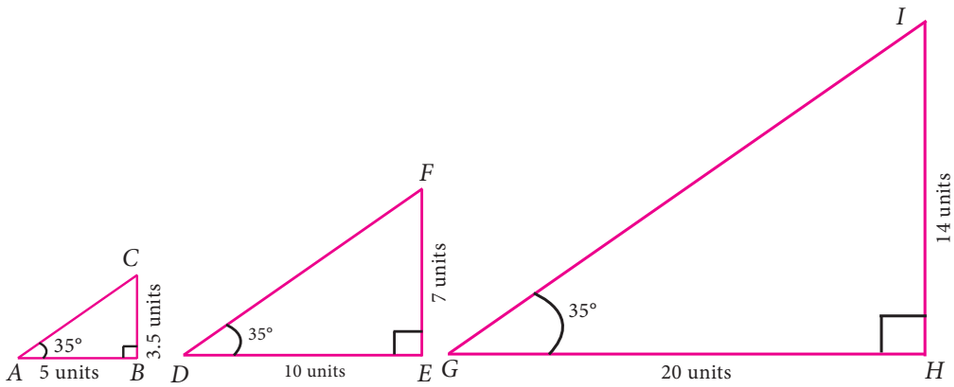


Fig. 6.1

What can you say about the ratio $\left(\frac{opposite\ side}{adjacent\ side}\right)$ in each case? Every right angled triangle given here has the same ratio 0.7 ; based on this finding, now what could be the length of the side marked '*x*' in the Fig 6.2? Is it 15?

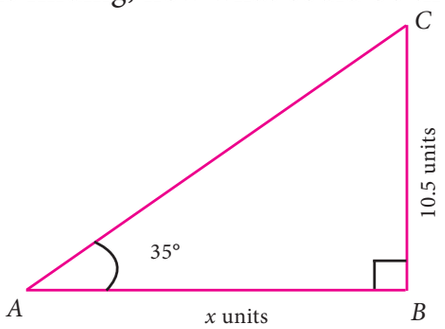
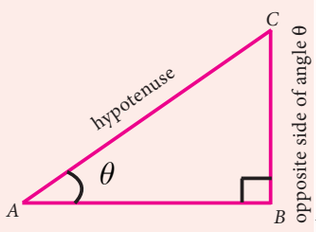
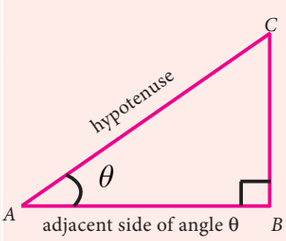
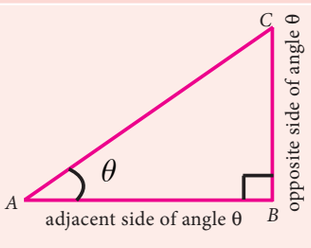


Fig. 6.2

Such remarkable ratios stunned early mathematicians and paved the way for the subject of trigonometry.

There are three basic ratios in trigonometry, each of which is one side of a right-angled triangle divided by another.

The three ratios are:

Name of the angle	sine	cosine	tangent
Short form	sin	cos	tan
Related measurements			
Relationship	$\sin\theta = \frac{opposite\ side}{hypotenuse}$	$\cos\theta = \frac{adjacent\ side}{hypotenuse}$	$\tan\theta = \frac{opposite\ side}{adjacent\ side}$

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